

● RESEARCH GRADE · 99%+ PURITY · USA-BASED

RESEARCH REFERENCE & SAFETY DOCUMENTATION

Selank

Synthetic tuftsin analogue heptapeptide · Research-grade reference material

SEQUENCE
TKPRPGP

CAS
129954-34-3

FORMULA
 $C_{33}H_{57}N_{11}O_9$

MW
751.9 g/mol

Sold for laboratory research purposes only. Not for human or veterinary use, consumption, or therapeutic application. Must be 21+ to purchase. | Independently verified · HPLC + mass spectrometry · Certificate of Analysis per lot.

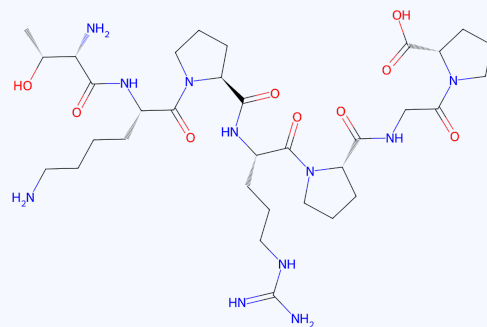
Identity & Composition

Selank is a synthetic heptapeptide based on tuftsin — an endogenous immunomodulatory tetrapeptide (Thr-Lys-Pro-Arg) produced by enzymatic cleavage of immunoglobulin G — extended with a C-terminal Pro-Gly-Pro tripeptide that increases metabolic stability. Developed at the Institute of Molecular Genetics, Russian Academy of Sciences.

Product name	Selank (TP-7)
Class	Synthetic tuftsin analogue
Sequence	Thr-Lys-Pro-Arg-Pro-Gly-Pro
Parent peptide	Tuftsin (Thr-Lys-Pro-Arg)
CAS number	129954-34-3
Molecular formula	C33H57N11O9
Molecular weight	751.9 g/mol
Appearance	White lyophilized powder
Identified use	In-vitro research & development only

Thr Lys Pro Arg Pro Gly Pro

Primary sequence (N → C): TKPRPGP



Molecular structure. 2D structure computed from the primary sequence; formula and mass consistent with C33H57N11O9 (751.9 g/mol). The C-terminal Pro-Gly-Pro extension confers resistance to enzymatic degradation relative to tuftsin.

Shared research lineage. Selank originates from the same institution as Semax and shares its design logic — an endogenous peptide fragment stabilized by a Pro-Gly-Pro C-terminus. The two are distinct compounds with different parent molecules and different studied targets.

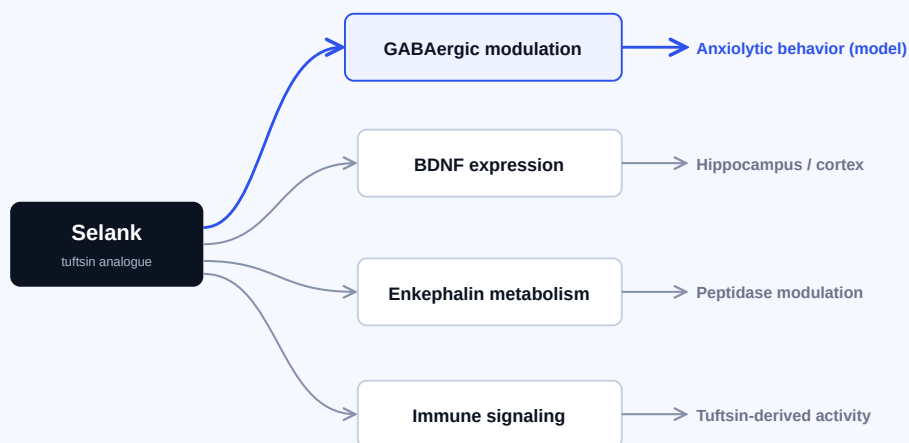
Research Background

Selank has been studied predominantly by Russian and Eastern European research groups since the 1990s, in anxiety, cognition and immune-signaling models. Its tuftsin origin gives it a dual research profile spanning both CNS and peripheral immune activity — unusual among neuroactive peptides.

Mechanisms studied in preclinical models

Findings below derive from animal and cell-culture systems. They are not established effects in humans.

- **GABAergic modulation.** Described in the literature as acting on GABA binding / receptor affinity rather than as a direct receptor agonist.
- **BDNF regulation.** Kolik et al. (2019) reported Selank (0.3 mg/kg IP, 7 days) prevented memory and attention impairment during alcohol withdrawal in rats and prevented ethanol-induced BDNF elevation in hippocampus and prefrontal cortex.
- **Gene-expression effects.** Kolomin et al. (2010) reported a single intranasal dose produced measurable gene-expression changes in both rat hippocampus *and* spleen — consistent with dual CNS/immune activity.
- **Behavioral models.** Seregin et al. reported anxiolytic activity in elevated plus-maze models without motor impairment or sedation.
- **Monoaminergic & enkephalin systems.** Reported modulation of serotonergic signaling and enkephalin metabolism.



Proposed pathways investigated in preclinical (animal / in-vitro) models. Schematic only; not a claim of established human effect.

What the Literature Supports — and What It Doesn't

Selank follows the same evidentiary pattern as Semax: a substantial preclinical record and a national regulatory approval, alongside an absence of independent Western replication. Both facts belong in an honest reference.

Evidence at a glance

PRECLINICAL

Substantial

WESTERN RCT

None replicating

RUSSIA

Approved anxiolytic

US FDA

Not approved

- Russian clinical reports describe anxiolytic activity in generalized anxiety disorder without sedation; **numeric effect sizes are frequently absent from English-language abstracts**, limiting independent appraisal.
- No Western randomized controlled trials have replicated** the reported clinical findings.
- Preclinical findings are specific and quantified (dose, duration, statistical significance), which is a genuine strength of the animal literature.
- Comparisons to benzodiazepines appear in the Russian literature; these should not be read as establishing equivalence under Western regulatory standards.

Appropriate characterization: a metabolically stabilized tuftsin analogue with a coherent preclinical GABAergic/BDNF profile and a national regulatory approval, lacking the independent controlled human evidence required for FDA authorization.

Regulatory & status notes

US FDA	Not approved for any use.
Russia	Reported as an approved prescription anxiolytic. Approval in one jurisdiction does not imply FDA/EMA acceptance.
Supply basis	Research reagent for in-vitro laboratory use only; not for human or veterinary administration.

Selected references

- Kolik LG, et al. Selank, Peptide Analogue of Tuftsin, Protects Against Ethanol-Induced Memory Impairment by Regulating BDNF Content in the Hippocampus and Prefrontal Cortex in Rats. Bull Exp Biol Med. 2019.
- Kolomin T, et al. Changes in gene expression in rat hippocampus and spleen following a single intranasal dose of Selank. Dokl Biochem Biophys. 2010.
- Seregin et al. Anxiolytic activity of Selank in elevated plus-maze models. PMID 19916388.
- Teleshova ES, et al. Results of clinical and pharmacological studies of the peptide anxiolytic Selank. Psikhatriya. 2010;(4):26–35. [Russian]

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	Selank
CAS	129954-34-3
Sequence	THR - LYS - PRO - ARG - PRO - GLY - PRO
Formula / MW	C33H57N11O9 / 751.9
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use
Supplier	
Emergency	

2 Hazard Identification

Not classified based on currently available data. Limited toxicological data; absence of classification is not a determination of absence of hazard. **Signal word:** None. **Pictograms:** None required. **Precautionary:** P261, P264, P280, P501. Handle as a potentially bioactive peptide of unknown toxicity.

3 Composition

Single-substance product, >98% research grade. Residual synthesis reagents, solvents and counter-ions may be present consistent with research-grade material. See lot COA for impurity profile.

4 First-Aid Measures

Eyes: rinse with water. **Skin:** wash with soap and water; remove contaminated clothing. **Inhalation:** fresh air. **Ingestion:** rinse mouth; do not induce vomiting unless directed. Treat symptomatically; no specific antidote known.

5 Fire-Fighting

Media: CO₂, dry chemical, foam, water spray. Combustion may produce CO, CO₂, NO_x and sulfur oxides. Use SCBA and full protective gear.

ANVIL COMPOUNDS

6 Accidental Release

Avoid dust; ventilate; use PPE per §8. Prevent entry to drains/waterways. Collect in a closed labeled container for disposal per §13.

7 Handling & Storage

Handle as bioactive peptide of unknown toxicity; minimize exposure. Weigh/reconstitute in fume hood. Store lyophilized solid tightly closed, protected from light/moisture/heat; -20 °C long-term, 2-8 °C short-term. Aliquot reconstituted solutions; store frozen, protected from light. Observe COA retest date.

8 Exposure Controls / PPE

No established OEL; control to ALARA. Fume hood for dry powder. Nitrile gloves (double-glove for neat powder); ANSI Z87.1 eye protection; lab coat; N100/P100 respirator where controls insufficient. Eyewash/safety shower per ANSI Z358.1.

9 Physical & Chemical Properties

State	Lyophilized powder
Appearance	White powder
Odor	Odorless
Solubility	Soluble in water
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable under normal storage. Avoid heat, moisture, sunlight, UV, open flame. Incompatible with strong oxidizers, strong acids/bases, strong reducing agents. Decomposition: CO, CO₂, NO_x, SO_x. No hazardous polymerization.

11 Toxicological Information

Not fully characterized as a research chemical. No LD₅₀/LC₅₀ available from authoritative databases. Skin/eye: treat as potentially irritating. Sensitization: cannot be ruled out. Mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: no authoritative substance-specific data; not classified based on available data; hazards cannot be excluded. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

No substance-specific ecotoxicity, persistence or bioaccumulation data identified. Avoid release to surface water, soil, drains or sewers.

13 Disposal

Dispose per applicable local/state/federal regulations; US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

14 Transport Information

Not regulated as dangerous goods under DOT (49 CFR), IATA or IMDG based on current classification. UN number: not applicable. Shipper must independently verify.

15 Regulatory Information

US TSCA: may qualify for R&D exemption (40 CFR 720.36). OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for SR&D use; verify exemption. Not a legal determination of regulatory status.

16 Other Information

Revision [DATE] · Version 1.0. Independent professional review recommended prior to use. Sources: PubChem; peer-reviewed literature; OSHA HazCom 2012; GHS Rev. 8; OECD Test Guidelines. Provided without warranty. For scientific R&D use only.

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